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Teo cociente dalembert

Teorema 1 (Criterio del cociente de D'Alembert).

Sea $(z_n)_n \subset \mathbb{C}$ *tal que* $\forall n \in \mathbb{N} \cup \{0\} : z_n \neq 0$ *y* $\lim_{n\to\infty} \frac{|z_{n+1}|}{|z_n|} = \lambda \in [0, \infty]$, *entonces*

(1) $\lambda < 1 \implies \sum z_n$ *converge absolutamente.*

(2) $\lambda > 1 \implies \sum z_n$ *diverge.*

***Demostración*:** Evaluemos por casos según el valor de λ:

(1) Si $\lambda \in [0, 1)$, definimos $r = \frac{\lambda+1}{2}$ tal que $\lambda < r < 1$. Tomamos $\varepsilon = r - \lambda$, $\exists N_0 \in \mathbb{N}$:

$$\forall n \geqslant N_0 : \left| \frac{|z_{n+1}|}{|z_n|} - \lambda \right| < \varepsilon \implies \frac{|z_{n+1}|}{|z_n|} < \varepsilon + \lambda = r \implies |z_{n+1}| < r\,|z_n|\,.$$

Por tanto, $|z_{N_0+1}| < r\,|z_{N_0}| \implies |z_{N_0+2}| < r^2\,|z_{N_0}| \implies \forall k \in \mathbb{N} : |z_{N_0+k}| < r^k\,|z_{N_0}|$.

$$\implies \sum_{n=N_0}^{\infty} |z_n| = \sum_{k=0}^{\infty} |z_{N_0+k}| < |z_{N_0}| \sum_{k=0}^{\infty} r^k \text{ que converge.}$$

Entonces, por el corolario del criterio de Cauchy, $\sum_n |z_n|$ converge, luego $\sum_n z_n$ converge absolutamente.

(2) Si $\lambda \in (1, \infty)$, definimos $r = \frac{\lambda+1}{2}$ tal que $1 < r < \lambda$. Tomamos $\varepsilon = \lambda - r$, $\exists N_0 \in \mathbb{N}$:

$$\forall n \geqslant N_0 : \left| \frac{|z_{n+1}|}{|z_n|} - \lambda \right| < \varepsilon \implies \frac{|z_{n+1}|}{|z_n|} > \lambda - \varepsilon = r \implies |z_{n+1}| > r\,|z_n|\,.$$

Por tanto, $|z_{N_0+1}| > r\,|z_{N_0}| \implies |z_{N_0+2}| > r^2\,|z_{N_0}| \implies \forall k \in \mathbb{N} : |z_{N_0+k}| > r^k\,|z_{N_0}|$. Entonces, $\lim_{n\to\infty} |z_n| \neq 0$, luego $\lim_{n\to\infty} z_n \neq 0$ y la serie $\sum z_n$ diverge.

(3) Si $\lambda = \infty$ entonces $\forall M > 0 : \exists N_0 \in \mathbb{N} : \forall n \geqslant N_0 : |z_n| > M\,|z_{N_0}|$. Entonces, $|z_n| \nrightarrow 0$ y la serie $\sum z_n$ diverge (por el `teo-convergencia-serie-imp-lim-0/`**??**).

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Ejemplos 1 (Si $\boldsymbol{\lambda = 1}$, el criterio del cociente no es concluyente).

$\boxed{1}$ Si $\sum a_n$ está dada por $\forall n \in \mathbb{N} : a_n = \frac{1}{n}$, entonces, $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = 1$ y $\sum a_n$ diverge.

$\boxed{2}$ Si $\sum b_n$ está dada por $\forall n \in \mathbb{N} : b_n = \frac{1}{n^2}$, entonces, $\lim\limits_{n\to\infty} \dfrac{b_{n+1}}{b_n} = 1$ y $\sum b_n$ converge.

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